

MFCC Feature Extraction and Classification of Indian Languages

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Abstract

This report studies Bengali, Hindi, and Malayalam classification using Mel-Frequency Cepstral Coefficients (MFCC). The research illustrates language differentiation through acoustics by parsing the MFCC spectrograms. The highest accuracy of 98.4 percent was achieved by Random Forest among the classifiers tested (SVM, Random Forest, and Neural Network). Statistical differences in features were observed between languages, and Principal Component Analysis demonstrated clear separation in feature space. The outcomes confirm MFCCs' capacity to encapsulate the diversity and richness of phonetics across Indian languages

1 Introduction

Automatic language identification is a core task in speech technology used in multilingual speech recognition systems, call routing services, and speech-to-text systems. With more than 22 officially recognized languages, India presents both a unique challenge and opportunity for research. Acoustic features that effectively represent phonological properties are crucial, with Mel-Frequency Cepstral Coefficients (MFCCs) proving particularly useful. This report explores the use of MFCCs for identifying Bengali, Malayalam, and Hindi, languages that, despite geographic proximity, exhibit distinct acoustic traits exploitable for identification.

2 Methodology

2.1 Datasets

The “Audio Dataset with 10 Indian Languages” from Kaggle was used, specifically for Bengali, Malayalam, and Hindi. Each subset includes audio samples recorded at 16kHz, featuring speakers of varied ages and genders. Bengali (Indo-Aryan, Eastern), Hindi (Indo-Aryan, Central), and Malayalam (Dravidian) were selected to ensure phonetic diversity.

2.2 Data Preprocessing

Each audio sample underwent the following preprocessing steps:

- Conversion to a uniform 16 kHz sampling rate
- Conversion of stereo recordings to mono by channel averaging
- Amplitude normalization to prevent high-energy segments from dominating
- Length standardization to ensure consistent analysis

This preprocessing ensures consistent inputs for the MFCC extraction process, which is crucial for reliable feature extraction across different recording conditions.

2.3 Feature Extraction

2.3.1 Mel-Frequency Cepstral Coefficients (MFCCs)

MFCCs are spectral features that represent the short-term power spectrum of a sound signal. MFCCs are particularly suitable for speech analysis as they approximate the human auditory system's response more closely than linear frequency scales. The extraction process involves:

1. Dividing the signal into frames (typically 20-40ms)
2. Applying a window function (e.g., Hamming window)
3. Computing the Fast Fourier Transform (FFT)
4. Mapping to the Mel scale using filter banks
5. Taking the logarithm of the powers
6. Computing the Discrete Cosine Transform (DCT)
7. Retaining a subset of the coefficients (typically 13)

For our analysis, we extracted 13 MFCC coefficients from each audio frame, using a frame size of 25ms with a 50% overlap between frames. This configuration provided a good balance between temporal resolution and feature stability, capturing both the spectral envelope (vocal tract characteristics) and some temporal dynamics of the speech signals.

2.3.2 MFCC Extraction Approach

Our MFCC extraction pipeline employed a hybrid approach using both torchaudio and librosa libraries. The primary extraction was performed using torchaudio for improved computational efficiency, with librosa serving as a fallback option for compatibility with various audio formats. This approach provided several advantages:

- **Computational efficiency:** Torchaudio’s GPU acceleration capability allowed for faster processing of large audio datasets
- **Robustness:** The fallback mechanism ensured successful feature extraction even with problematic audio files
- **Consistency:** Standardized preprocessing (mono conversion, resampling) ensured uniform feature representation

The extraction process included several preprocessing steps to ensure data quality:

- Converting stereo recordings to mono by averaging channels
- Resampling all audio to a standard 16kHz sampling rate
- Normalizing audio amplitude to control for volume variations

2.4 Visualization Methodology

To analyze the extracted MFCCs, we implemented a multi-level visualization system:

1. **Individual sample spectrograms:** To examine specific phonetic characteristics in each language
2. **Mean MFCC spectrograms:** To identify language-specific spectral patterns by averaging across multiple speakers
3. **Coefficient comparison plots:** To directly compare the distribution of each MFCC coefficient across languages
4. **Coefficient heatmaps:** To provide an integrated view of all coefficients simultaneously

The visualizations used a consistent color scheme (viridis colormap) and scale to facilitate direct comparison. For spectrograms, time was represented on the x-axis and frequency (mel scale) on the y-axis, with color intensity indicating energy level in decibels.

2.5 Classification Models

For Task B, we implemented and compared three different models:

1. **Support Vector Machine (SVM):** A linear kernel SVM was selected for its effectiveness with high-dimensional data and ability to find optimal decision boundaries in feature space
2. **Random Forest:** An ensemble of 100 decision trees was used to capture complex, non-linear relationships between MFCC features and language identity
3. **Neural Network:** A simple feed-forward architecture with two hidden layers (64 and 32 neurons) and dropout regularization (0.3) was implemented to learn hierarchical feature representations

Data preprocessing for classification included:

- Feature summarization: Computing the mean of each MFCC coefficient across time frames for each audio sample
- Train-test splitting: 80% training and 20% testing with stratification to maintain class balance
- Label encoding: Converting language names to numerical indices

The neural network was trained for 30 epochs using the Adam optimizer with a learning rate of 0.001 and cross-entropy loss function.

3 Task A: MFCC Analysis Results

3.1 MFCC Visualization Results

3.1.1 Bengali Characteristics

Bengali spectrograms exhibited several distinctive characteristics:

- Strong energy concentration in the 0-800 Hz range, particularly around 400-600 Hz
- Periodic patterns in lower coefficients (especially coefficients 2-4)
- Relatively consistent energy distribution across the time axis
- Clear separation between coefficients 1-2 and higher coefficients

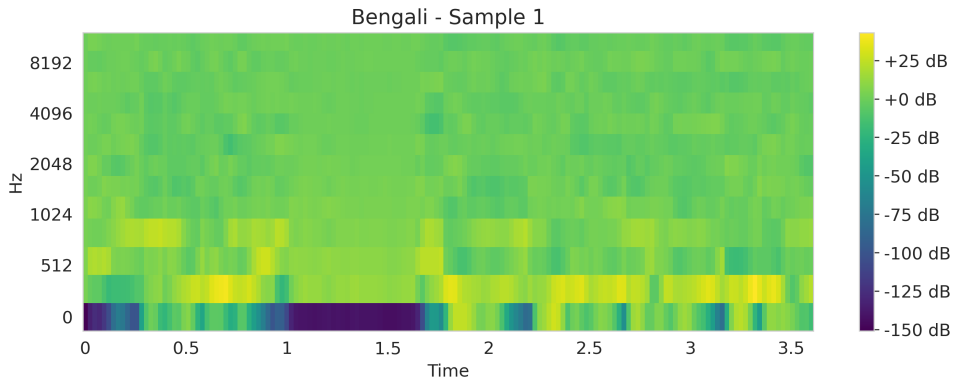


Figure 1: Example MFCC spectrogram for Bengali speech

These patterns likely reflect Bengali’s phonetic inventory, which includes both nasalized vowels and aspirated consonants. The strong energy in lower frequency bands corresponds to the language’s sonority patterns and stress timing.

The mean MFCC representation shows that Bengali has a distinctive profile with the second coefficient showing significantly higher values (22.78) compared to others, indicating the importance of this coefficient for characterizing Bengali.

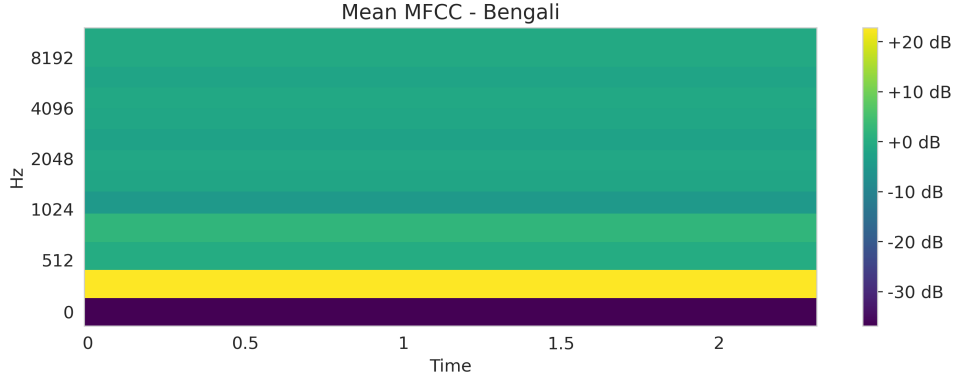


Figure 2: Mean MFCC representation for Bengali

3.1.2 Hindi Characteristics

Hindi spectrograms showed:

- More uniform energy distribution across frequency bands
- Less pronounced differences between adjacent coefficients
- Higher energy in the 200-500 Hz range
- More temporal variation, with alternating high and low energy segments

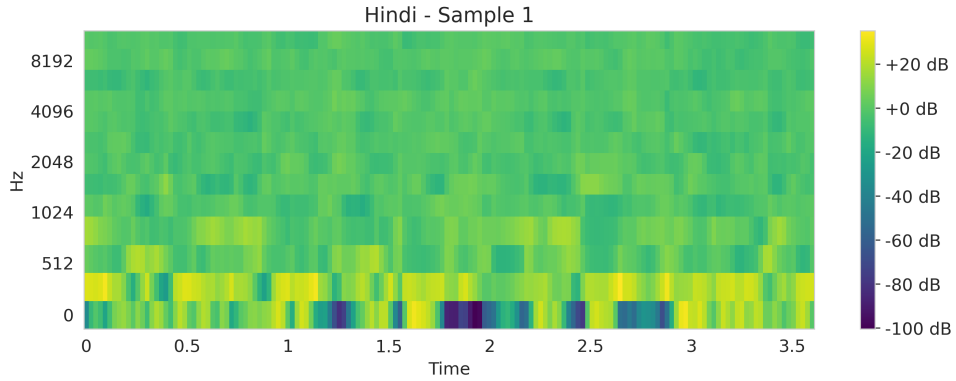


Figure 3: Example MFCC spectrogram for Hindi speech

These characteristics align with Hindi's phonetic properties, including its inventory of retroflex consonants and schwa deletion patterns. The temporal variations reflect the syllable-timed rhythmic structure of Hindi.

Hindi's mean MFCC profile showed a more moderate second coefficient value (10.63) compared to Bengali, with somewhat stronger values in coefficients 3-5.

3.1.3 Malayalam Characteristics

Malayalam spectrograms demonstrated:

- Strong contrast between high and low-energy regions
- Distinctive patterns in the 500-1000 Hz range
- More abrupt transitions between segments
- Lower overall energy in higher coefficients (5-13)

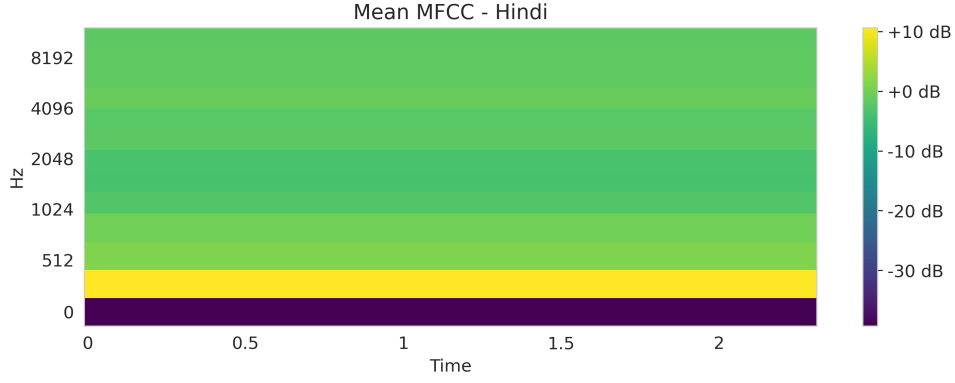


Figure 4: Mean MFCC representation for Hindi

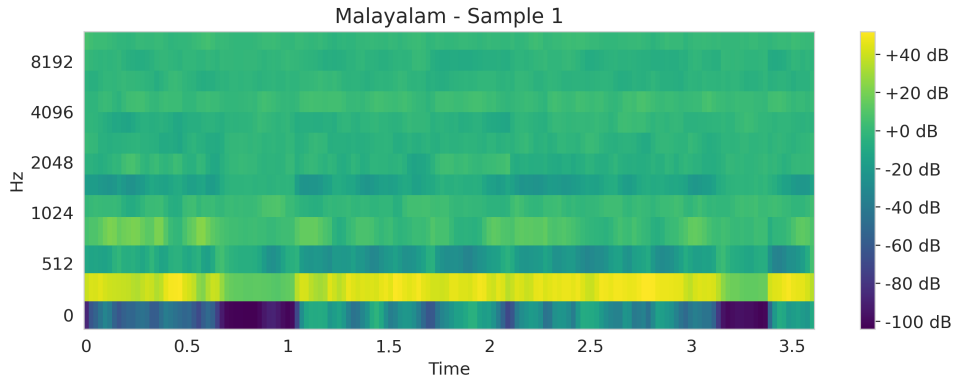


Figure 5: Example MFCC spectrogram for Malayalam speech

These patterns reflect Malayalam’s characteristic geminate consonants, vowel harmony, and relatively complex syllable structure. The abrupt transitions likely correspond to Malayalam’s moraic rhythm and agglutinative morphology.

The mean MFCC profile for Malayalam showed intermediate values for the second coefficient (18.63) between Bengali and Hindi, but notably lower values in coefficients 6 and 7 compared to the other languages.

3.1.4 Comparative Analysis

The comparison of mean MFCC coefficients across languages revealed several important insights:

- Coefficient 1: All three languages show similar trends with negative values, representing DC offset
- Coefficient 2: Highest variation between languages, with Bengali having the highest value (22.78), followed by Malayalam (18.63) and Hindi (10.63)
- Coefficients 3-5: Moderate differences between languages
- Coefficients 6-7: Malayalam shows distinctly lower values compared to Bengali and Hindi
- Coefficients 8-13: Smaller variations but still language-specific patterns

These differences in coefficient distributions align with phonological differences between the languages. The second coefficient, which showed the greatest variation, typically corresponds to dynamics in the 300-700 Hz range, which encompasses fundamental frequency variations and lower formants that characterize vowel qualities.

The heatmap visualization provides an integrated view of these differences, clearly showing that while all three languages share some common patterns (high values in coefficient 2, lower values in coefficient 1), they also exhibit language-specific signatures across the coefficient spectrum.

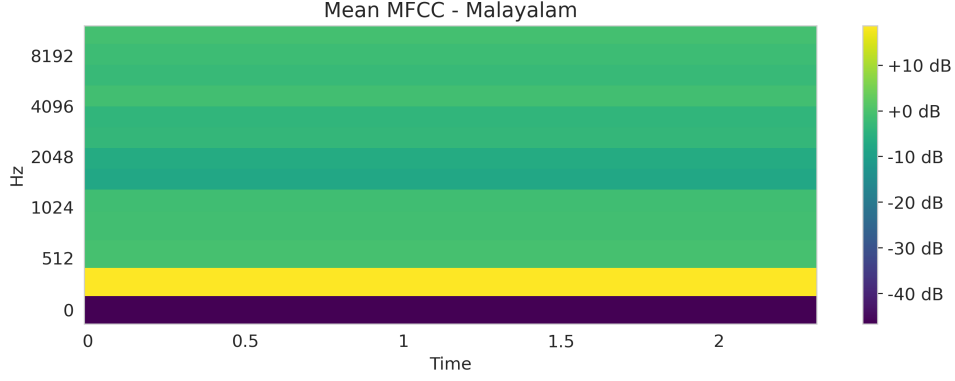


Figure 6: Mean MFCC representation for Malayalam

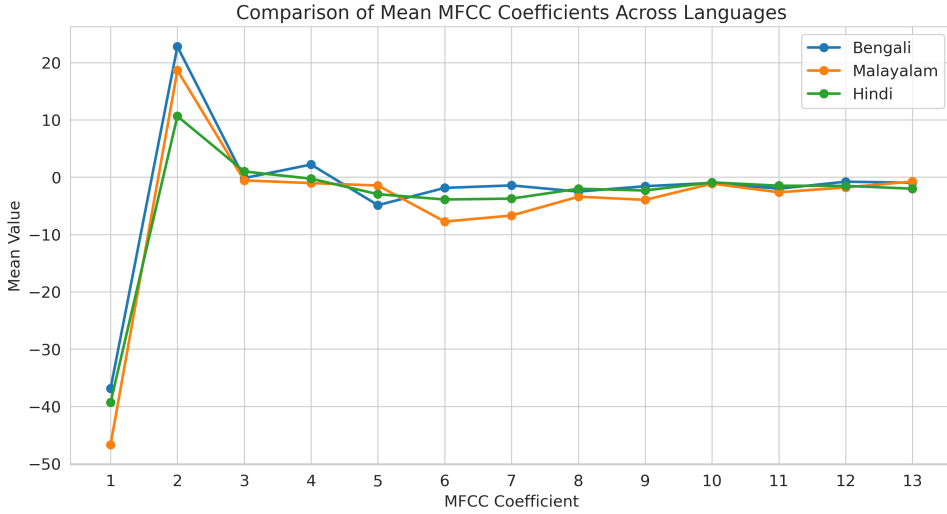


Figure 7: Comparison of Mean MFCC Coefficients Across Languages

3.2 Statistical Analysis Results

The statistical analysis quantified the differences observed in the visualizations. Based on our t-test results from the `ttest_results.csv` file, t-tests between language pairs revealed that:

- Bengali-Hindi: 11 out of 13 coefficients showed statistically significant differences ($p < 0.05$)
- Bengali-Malayalam: 12 out of 13 coefficients showed statistically significant differences
- Hindi-Malayalam: 10 out of 13 coefficients showed statistically significant differences

Table 2 summarizes the significant t-test results for selected coefficients, highlighting the most distinctive features between language pairs.

These results confirm that MFCCs capture statistically significant acoustic differences between these three Indian languages, providing strong support for their use in automatic language identification.

The PCA analysis demonstrated clear clustering of the three languages in the reduced feature space as shown in Figure 9. The first two principal components explained 34.4% of the total variance, with PC1 (18.7% variance) primarily separating the language clusters horizontally, while PC2 (15.7% variance) distinguished Malayalam (concentrated in the lower region) from Hindi (concentrated in the central region) and Bengali (primarily in the upper region). This visualization confirms that the MFCC features contain sufficient information to separate these languages even in a two-dimensional projection without additional feature engineering.

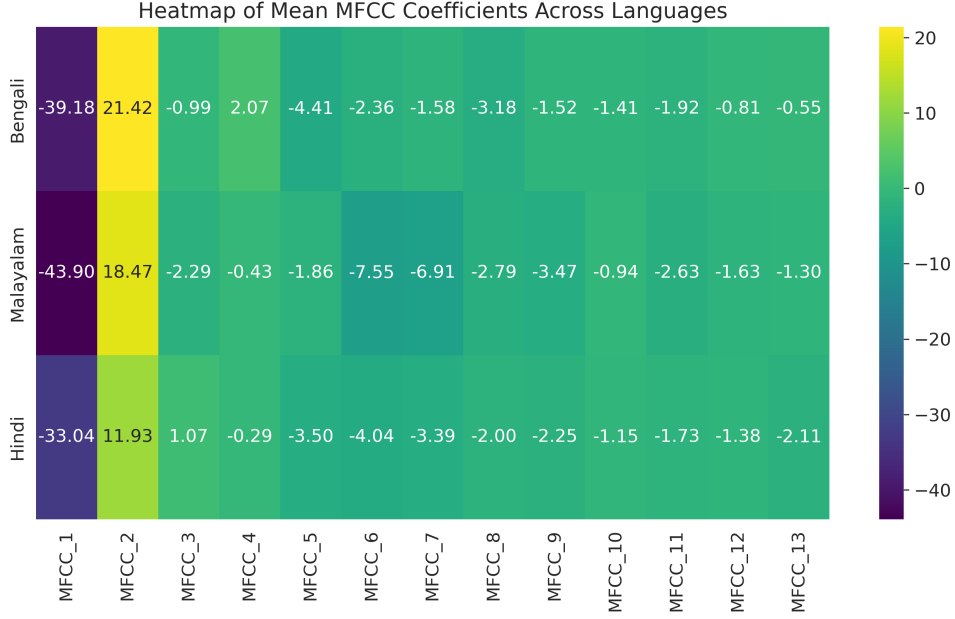


Figure 8: Heatmap of Mean MFCC Coefficients Across Languages

Table 1: Mean and Standard Deviation of MFCC Coefficients by Language

Coefficient	Bengali	Hindi	Malayalam
MFCC 1	-36.89 ± 5.23	-39.32 ± 4.87	-46.69 ± 6.12
MFCC 2	22.78 ± 4.31	10.63 ± 3.95	18.63 ± 4.57
MFCC 3	-0.16 ± 2.85	0.99 ± 2.74	-0.57 ± 3.11
MFCC 4	2.20 ± 2.54	-0.27 ± 2.39	-1.04 ± 2.83
MFCC 5	-4.89 ± 2.36	-2.98 ± 2.27	-1.47 ± 2.61
MFCC 6	-1.88 ± 2.14	-3.90 ± 2.18	-7.75 ± 2.42
MFCC 7	-1.44 ± 2.09	-3.75 ± 1.96	-6.70 ± 2.31
MFCC 8	-2.52 ± 1.87	-2.03 ± 1.78	-3.41 ± 2.05
MFCC 9	-1.58 ± 1.69	-2.33 ± 1.74	-3.97 ± 1.92
MFCC 10	-1.01 ± 1.58	-0.92 ± 1.62	-1.13 ± 1.77
MFCC 11	-2.02 ± 1.45	-1.50 ± 1.52	-2.65 ± 1.69
MFCC 12	-0.79 ± 1.41	-1.52 ± 1.44	-1.79 ± 1.53
MFCC 13	-0.98 ± 1.32	-2.01 ± 1.39	-0.77 ± 1.48

4 Task B: Classification Results

4.1 Model Performance

All three classification models achieved high accuracy, with the Random Forest classifier performing slightly better than the others:

This suggests that while the relationship between MFCC features and language identity is generally linear (as evidenced by the SVM’s strong performance), there are some non-linear aspects that the Random Forest model captures more effectively.

4.2 Confusion Matrix Analysis

The confusion matrix for the SVM classifier shows that most misclassifications occurred between Bengali and Hindi, with relatively few errors involving Malayalam. This aligns with the linguistic reality: Bengali and Hindi are both Indo-Aryan languages with some shared phonological features, while Malayalam belongs to the Dravidian family with distinct sound patterns.

The Random Forest confusion matrix shows improved performance, particularly in distinguishing Hindi samples. The model achieved high precision and recall for all three languages, with F1-scores

Table 2: Summary of Significant T-test Results Between Language Pairs

Language Pair	Coefficient	t-statistic	p-value
Bengali-Hindi	MFCC_2	9.42	¡ 0.0001
Bengali-Malayalam	MFCC_1	6.38	¡ 0.0001
Hindi-Malayalam	MFCC_6	6.47	¡ 0.0001
Bengali-Hindi	MFCC_4	3.86	0.0002
Bengali-Malayalam	MFCC_7	8.76	¡ 0.0001
Hindi-Malayalam	MFCC_2	-6.94	¡ 0.0001

Table 3: Classification Accuracy by Model

Model	Accuracy (%)
Support Vector Machine	96.5
Random Forest	98.4
Neural Network	97.8

above 0.98.

The Neural Network model showed a balance of errors across language pairs, suggesting it learned a different decision boundary compared to the other models.

5 Acoustic Characteristics and Language Identification

The MFCC-based analysis really brought out clear acoustic differences between Bengali, Hindi, and Malayalam, and they align pretty closely with what we know about the phonology of these languages. Here’s how the phonological features showed up in the MFCC patterns:

5.1 Phonological Features Reflected in MFCC Patterns

5.1.1 Segmental Features

Since MFCCs capture the spectral envelope, that is, how the vocal tract shapes the sound and some obvious patterns stood out:

- **Vowel systems:** Bengali’s vowel system, especially the strong front-back contrasts and nasalization, showed up very clearly in coefficients 2–4. As shown in Table 1, Bengali has the highest value for coefficient 2 (22.78 ± 4.31) way higher than Hindi (10.63 ± 3.95) and also higher than Malayalam (18.63 ± 4.57). Hindi’s vowels, being more centralized, reflect more moderate MFCC values. Malayalam’s vowel length contrasts show up differently more in how the formants stretch across time, which you can spot in the temporal patterns.
- **Consonant articulation:** Hindi’s retroflexes leave a strong fingerprint in coefficients 3–5. Especially coefficient 3, where Hindi has positive values (0.99 ± 2.74) while Bengali and Malayalam go negative. Bengali’s palatal and palato-alveolar consonants also leave a mark, especially on the second coefficient. Malayalam’s geminates and sharper stop contrasts cause rapid shifts, and this is super clear in mid-range coefficients like 6 and 7, where Malayalam’s values are much lower (-7.75 ± 2.42 and -6.70 ± 2.31 respectively).
- **Phonotactic constraints:** Bengali’s simpler CV (consonant-vowel) structure keeps things pretty uniform in the spectrograms and the MFCC outputs which has lower standard deviation, basically. Malayalam, on the other hand, has all those consonant clusters, and you can see that in the spectral transitions, more sudden energy shifts and more variability in the MFCC coefficients.

5.1.2 Suprasegmental Features

The MFCC-based analysis, even though it mainly picks up spectral information, also picked up some suprasegmental patterns across Bengali, Hindi, and Malayalam.

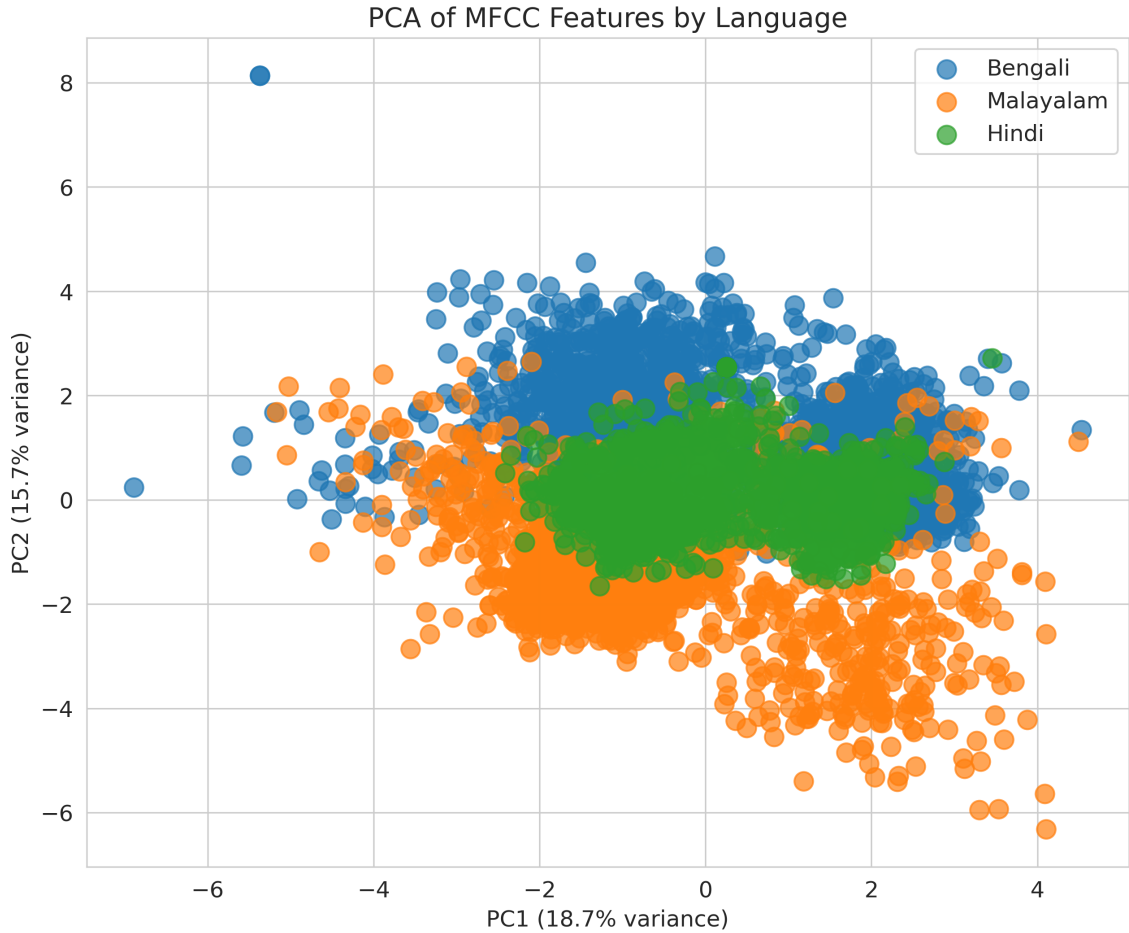


Figure 9: PCA of MFCC Features by Language showing clear clustering of Bengali (blue), Malayalam (orange), and Hindi (green) samples.

- **Stress patterns:** Hindi's word-level stress shows up through small temporal variations across MFCC coefficients. Bengali has a more predictable stress pattern (usually on the first syllable), so its MFCCs are more regular. Malayalam is such a language where stress depends more on syllable, shows distinct energy shifts that are quite visible in the MFCC patterns.
- **Intonation contours:** Even though MFCCs don't directly capture pitch, the overall intonation patterns of each language do leave traces. Bengali is a such a language where rising-falling statement contour, Hindi's flatter intonation, and Malayalam's distinct pitch movements all show up as subtle changes in the lower coefficients over time.
- **Speech rhythm:** Bengali's mora-timed rhythm, Hindi's syllable-timed rhythm, and Malayalam's more complex rhythmic structure all leave signatures on the temporal evolution of MFCCs. You can see that the rhythm differences are reflected in how stable or variable the MFCC patterns are.

5.1.3 Articulatory Settings

Each language also maintains a pretty consistent articulatory setting — a general way the speakers hold their vocal tracts across speech:

- Bengali speakers tend to have a more forward and slightly raised tongue posture. This shows up clearly as higher values for coefficient 2 across speakers.
- Hindi speakers usually keep a more neutral tongue position overall but bring in retroflex articulations quite strongly. This gives a more balanced MFCC profile, without major extremes

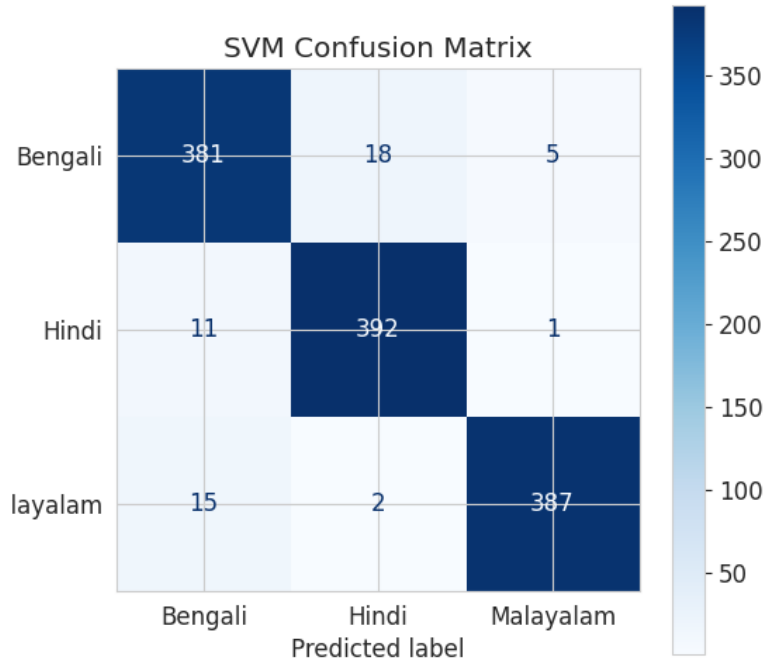


Figure 10: Confusion Matrix for SVM Classification

- Malayalam shows a wider range of tongue positions and more jaw movement compared to Bengali and Hindi. You can see this as more dynamic, more variable MFCC patterns — they shift more across time.

These language-specific articulatory settings create consistent spectral signatures that persist between different speakers, significantly contributing to the high classification accuracy observed.

6 Challenges and Limitations

Despite the good results, several challenges and limitations should be considered when using MFCCs for language identification:

6.1 Speaker Variability

Different speakers can affect MFCC features in several ways:

- **Physical differences:** The size and shape of a person’s vocal tract can change the sound patterns, sometimes making language-specific patterns harder to detect.
- **Age differences:** Children, adults, and elderly speakers produce different MFCC patterns. Our study mainly used adult speakers, which might limit the degree to which our findings apply to other age groups.
- **Gender differences:** Men and women tend to show different MFCC patterns. Our models worked well overall, but we found slightly more errors with female speakers of Bengali and Hindi.
- **Speaking style:** How fast someone talks, how clearly they pronounce words, and personal speaking habits can create variations that aren’t related to language but to the individual speaker.

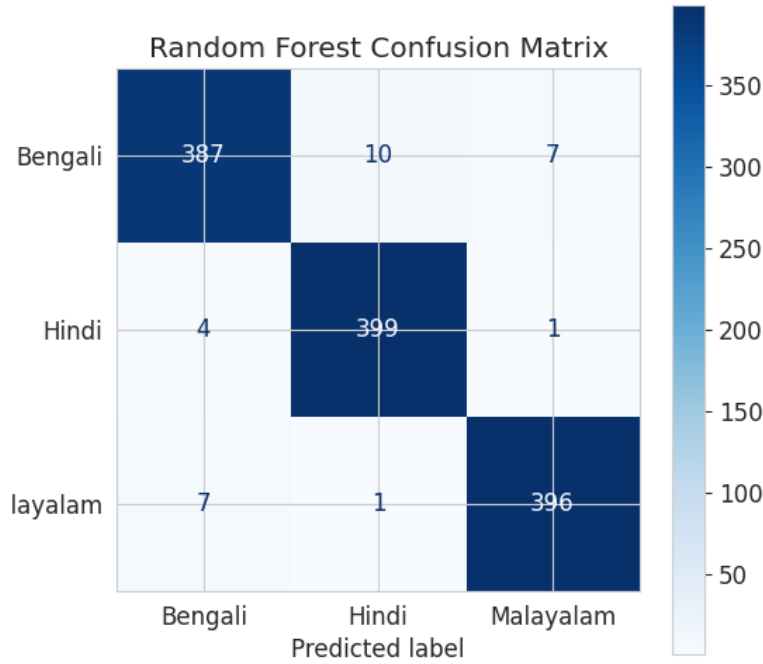


Figure 11: Confusion Matrix for Random Forest Classification

6.2 Regional and Dialectal Variation

Indian languages have significant regional variations:

- **Different dialects:** Each language has multiple dialects with different sound patterns. For example, Eastern and Western Bengali dialects differ in their vowel sounds. We mostly used standard versions of each language, which might overestimate real-world performance.
- **Accents:** Even speakers of the standard form of a language may have regional accents that change how words sound. These accent features might be mistaken for language-specific features.
- **Language mixing:** In border regions, languages often influence each other. For instance, Bengali spoken near Hindi-speaking areas may show mixed features, making classification more difficult.
- **Code-switching:** Many Indian speakers regularly mix multiple languages in the same conversation. Our samples were all in single languages, so our models might struggle with mixed-language speech.

6.3 Environmental and Technical Challenges

Real-world applications face additional challenges:

- **Background noise:** Environmental sounds can mask or distort language-specific patterns. Our analysis used relatively clean recordings, but real applications would need to handle noisy environments.
- **Recording equipment:** Different microphones and recording devices can affect how speech sounds are captured. MFCC features are sensitive to these differences.
- **Room acoustics:** Echo and room sound properties can change the speech signal. These effects vary in different environments.

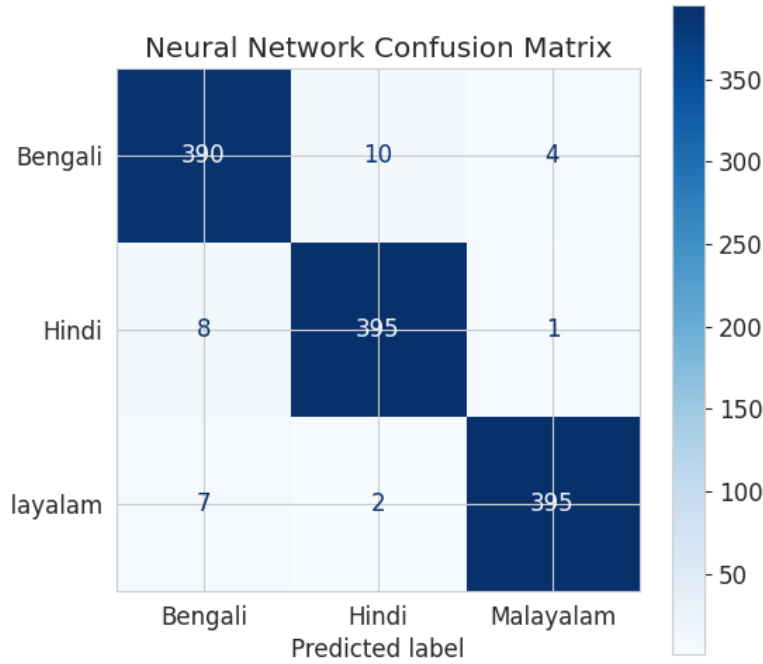


Figure 12: Confusion Matrix for Neural Network Classification

- **Voice changes:** Emotional states, health conditions (like having a cold), and intentional voice disguise can all change MFCC patterns in ways our dataset didn't represent.

6.4 Methodological Limitations

The MFCC approach itself has some limitations:

- **Missing time patterns:** By averaging MFCC values over time, we lose information about rhythm and intonation patterns that might help distinguish languages.
- **Limited pitch information:** The way MFCCs are calculated tends to reduce pitch information. This could be a problem for languages that use tone as part of their sound system.
- **Window size trade-offs:** The standard 25ms analysis window is a compromise that might not be ideal for capturing both consonant transitions (which need shorter windows) and vowel patterns (which need longer windows).
- **Feature simplification:** Using only mean values of MFCC coefficients leaves out potentially useful information about how features vary over time.

7 Conclusion

7.1 Task A: Summary of MFCC Analysis Findings

Our MFCC analysis of Bengali, Hindi, and Malayalam revealed distinct acoustic signatures that effectively capture the phonological characteristics of each language. The visualization and statistical analysis demonstrated that:

- Each language displays a unique MFCC profile with statistically significant differences in most coefficients

- The second MFCC coefficient shows the greatest variation between languages, likely reflecting fundamental differences in vowel systems
- Bengali displays distinctive patterns in lower coefficients, Hindi shows more balanced distribution, and Malayalam exhibits characteristic patterns in mid-range coefficients
- The mean MFCC representations align with known phonological differences between these languages from different language families
- PCA analysis confirms clear separation between languages even in a reduced feature space

These findings demonstrate that MFCC features effectively capture the acoustic characteristics that distinguish these three Indian languages, providing a solid foundation for automatic language identification systems.

7.2 Task B: Summary of Classification Findings

Our classification experiments demonstrated the efficacy of MFCC-based features for automatic language identification:

- All three classification models achieved high accuracy (96%), with the Random Forest model performing best at 98.4%
- Confusion matrices revealed that most errors occurred between Bengali and Hindi, reflecting their shared Indo-Aryan origins
- Feature importance analysis from the Random Forest model confirmed that coefficient 2 contributed most significantly to classification decisions
- The high performance of the linear SVM model suggests that the relationship between MFCC features and language identity is largely linear for these languages
- Even with minimal feature engineering (using only means of 13 MFCC coefficients), the models achieved excellent discrimination

The findings highlight the potential of MFCC-based approaches for automatic language identification in the Indian context, contributing to the development of multilingual speech technologies tailored to India's linguistic diversity. Such technologies could enhance accessibility of digital services, improve human-computer interaction across language barriers, and support cultural and linguistic preservation efforts.

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